Week 8 Material - Integration by Parts and Substitution

34. [1] Consider the finite region bounded by the functions $f(x) = \dfrac{x}{3}$ and $g(x) = \sqrt{x}$.

    (a) Sketch the graphs of $f(x)$ and $g(x)$, and then shade in the region defined by those boundaries. Clearly indicate the scale on both axes.

    (b) Write an integral that represents the area enclosed between these two graphs, computed using **vertical** slices, Δx or dx thick. You do **not** need to evaluate this integral.

    (c) Write an integral that represents the area enclosed between these two graphs, computed using **horizontal** slices, Δy or dy thick. You do **not** need to evaluate this integral.

35. [2] Evaluate the indefinite integral $\displaystyle\int xe^{-4x}\ \text{dx}$.

    Determine by differentiating whether or not your answer is correct.

36. [3] Evaluate the indefinite integral $\displaystyle\int \frac{2+x}{1+x^2}\ \text{dx}$.

    Determine by differentiating whether or not your answer is correct.

37. [4] Evaluate the following integrals:

    (a) $\displaystyle\int xe^{-3x^2}\ dx$

    (b) $\displaystyle\int \arccos(x)\ dx$

[1] Source: previous APSC 171 Test 2
[2] Source: previous APSC 171 Final Exam
[3] Source: previous APSC 171 Test 2
[4] Source: previous APSC 171 Test 2